1. Introduction

5. October 2026

1.1 A brief history of optimization

"Nothing takes places in the universe in which some rule of a maximum or minimum does not appear" - Euler, 1744

- 1939 Kantorowitsch, planning problem
- 1941 Hitchcock, transportation
- 1945 Stigler, diet problem
- 1945 Dantzig, simplex "
- 1940s Koopmans, economic modeling
- 1958 Gomory, integer optimization

(- 1979 Khachiyan, ellipsoid method
  1984 Karmarkar, interior point method ...)

1.2 Mathematical modeling & examples

Three basic ingredients:

-) **Decision variable** (represents quantities that can be chosen)
-) **Objective function** (measures quality of decision)
-) **Constraints** (conditions on the decisions)

Mathematical model ("program")

maximize (or minimize) the objective function subject to (s.t.) constraints

$\max$ $\min$

We call this program a linear problem / program (LP) if all functions for constraints & objectives are linear

Example 1.1 (Linear optimization problem)

$$\max \quad 3x_1 + 2x_2 \quad \leftarrow \text{objective}$$

$$\text{s.t.} \quad \left.\begin{aligned} x_1 + x_2 &\leq 4 \\ 2x_1 + x_2 &\leq 5 \\ x_1, x_2 &\geq 0 \end{aligned}\right\} \text{constraints}$$

Example 1.2 (Nonlinear problems)

$$\min \quad \underbrace{(x_1 - 1)^2 + (x_2 - 2)^2}_{\text{nonlinear objective}}$$

$$\text{s.t.} \quad \left.\begin{aligned} x_1 + x_2 &\leq 4 \\ x_1, x_2 &\geq 0 \end{aligned}\right\} \text{linear constraints}$$

Applications:

- production planning
- transportation, routing, supply-chain mgmt
- workforce / timetable scheduling
- portfolio selection / financial planning
- energy systems
- ...

Example 1.3. This example is from Cottle and Thapa's textbook *Linear and Nonlinear Optimization* and was first published in *Scientific American* by R. G. Bland in 1981. The production of ale and beer should be optimized, where corn, hops, and malt are limited. All other ingredients can be considered plentiful. To produce a barrel of ale, 5 lb of corn, 4 oz of hops and 35 lb malt are needed, and the profit is \$13. To produce a barrel of beer, 15 lb of corn, 4 oz of hops and 20 lb malt are needed, and the profit is \$23. The stock includes 480 lb of corn, 160 oz of hops and 1,190 lb of malt.



↳ Decision variables: x_A ... amount of ale to produce
x_B ... —"— beer —"—

Objective: max profit

Constraints: use no more than 480 lbs corn
—"— 160 oz hops
1,190 lb malt

Model: max $13x_A + 23x_B$

s.t. $5x_A + 15x_B \leq 480$ (corn)

$4x_A + 4x_B \leq 160$ (hops)

$35x_A + 20x_B \leq 1{,}190$ (malt)

$x_A, x_B \geq 0$

Example 1.4 (Stigler's diet problem). For a moderately active man weighing 154 pounds (approximately 70 kg), which daily quantities of 77 available foods should be consumed in order to minimize cost while meeting the nine nutritional recommendations published by the National Research Council in 1943?

Nutrient	Recommended daily allowance
Energy	3,000 kcal
Protein	70 g
Calcium	0.8 g
Iron	12 mg
Vitamin A	5,000 IU
Vitamin B_1	1.8 mg
Vitamin B_2	2.7 mg
Nicotinic acid	18 mg
Vitamin C	75 mg

- x_j ... daily quantity of food j $(j = 1, \dots, 77)$, decision variable
- c_j ... cost per unit food j
- a_{ij} ... amount of nutrient i $(i = 1, \dots, 9)$
- r_i ... RDA of nutrient i

Model:

$$\begin{aligned}
\min \quad & c_1 x_1 + \dots + c_{77} x_{77} && \text{[objective function]} \\
\text{s.t.} \quad & a_{1,1} x_1 + \dots + a_{1,77} x_{77} \geq 3000 && \text{[calorie / energy constr.]} \\
& \vdots \qquad\qquad \vdots \\
& a_{9,1} x_1 + \dots + a_{9,77} x_{77} \geq 75 && \text{[vitamin C]} \\
& x_1, \dots, x_{77} \geq 0
\end{aligned}$$

Example 1.5 (Vehicle production planning). A manufacturer produces two types of vehicles: conventional vehicles with internal-combustion engines and battery-electric vehicles. The final stage of production consists of *assembly* and *inspection and testing*. The manufacturer has committed to producing at least 100 vehicles of each type per month. During this period, 8,000 labor-hours are available for assembly and 4,000 labor-hours are available for inspection and testing.

A conventional vehicle requires 12 hours of assembly and 5 hours of inspection and testing. An electric vehicle requires 14 hours of assembly and 8 hours of inspection and testing.

The unit profit is 4,000€ for each conventional vehicle and 6,000€ for each electric vehicle. How can the manufacturer maximize profit?

$$\max \quad 4{,}000\, x_1 + 6{,}000\, x_2$$

x_1 ... # conventional
x_2 ... # electric

$$\text{s.t.} \quad x_1 \geq 100$$
$$x_2 \geq 100$$
$$12x_1 + 14x_2 \leq 8000 \qquad \text{Assembly hours}$$
$$5x_1 + 8x_2 \leq 4{,}000 \qquad \text{Labor} \quad "$$
$$x_1, x_2 \in \mathbb{N}_0 = \{0, 1, 2, \dots\}$$

↳ This is an example of an integer optimization problem

Example 1.6 (Investment planning – knapsack problem). A company wants to open new shops in Austria and has seven potential sites for them. The site j has costs of c_j and projected profits of p_j:

j	Site	c_j in €	p_j in €
1	Wien	300	500
2	Graz	250	400
3	Linz	250	300
4	Salzburg	400	100
5	Innsbruck	350	400
6	Klagenfurt	200	600
7	Villach	150	500

The entire budget is 1,000 €, and the profit should be maximized. Which sites should the company open?

↳ $\max \quad 500x_1 + 400x_2 + 300x_3 + 100x_4 + 400x_5 + 600x_6 + 500x_7$

$\text{s.t.} \quad 300x_1 + 250x_2 + 250x_3 + 400x_4 + 350x_5 + 200x_6 + 150x_7 \leq 1000$

$x_1, \dots, x_7 \in \{0,1\}$

1.3 Notational conventions & mathematical foundations

Unless otherwise stated, all vectors are column vectors. An n-dim **vector** is ordered collection of n numbers:

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n$$

Transpose of x:

$$x^T = (x_1, x_2, \dots x_n) \in \mathbb{R}^{1 \times n}$$

$$(x^T)^T = x$$

For vectors $x, y \in \mathbb{R}^n$, the product

$$x^T y = \sum_{i=1}^{n} x_i y_i \in \mathbb{R}$$

is called the **inner product** (dot product)

Notation "$x \geq y$" is to be understood componentwise $(x_1 \geq y_1, \dots, x_n \geq y_n)$

Matrix:

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & & a_{mn} \end{pmatrix} \in \mathbb{R}^{m \times n} \quad \rightsquigarrow A^T \in \mathbb{R}^{n \times m}$$

Identity matrix

$$I_n = \begin{pmatrix} 1 & 0 & 0 & & 0 \\ 0 & 1 & 0 & & \\ & 0 & 1 & & \\ & & & & 1 \\ 0 & & & & 0 \\ & & & & 1 \end{pmatrix} \in \mathbb{R}^{n \times n} \quad \rightsquigarrow \begin{array}{l} I_n x = x \\ I_m A = A \\ A I_n = A \end{array}$$

Let $a_1, \dots, a_k \in \mathbb{R}^n$. The **span** of these vectors is the set of all linear combinations of them:

$$\operatorname{span}\{a_1, \dots, a_k\} = \left\{ \sum_{i=1}^{k} \lambda_i a_i \;\middle|\; \lambda_i \in \mathbb{R} \right\}$$

The vectors $a_1, \dots, a_k$ are **linearly independent** if

$$\sum_{i=1}^{k} \lambda_i a_i = 0 \Rightarrow \lambda_1 = 0, \dots, \lambda_k = 0$$

Otherwise, **linearly dependent**.

The dimension of $\operatorname{span}\{a_1, \dots, a_k\}$ ... maximum # of linearly independent vectors that can be selected from $a_1, \dots, a_k$

Row space $\operatorname{row}(A)$ is the span of its row vectors
Column " $\operatorname{col}(A)$ —— " column "

Rank of matrix A ...

$$\operatorname{rank}(A) = \dim(\operatorname{row}(A)) = \dim(\operatorname{col}(A))$$

Example 1.7 (Linear indep & rank)

$$A = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 2 & 3 & -1 & 0 \\ 1 & 2 & -1 & -1 \end{pmatrix} \quad \begin{matrix} r_1 = (1\ 1\ 0\ 1) \\ r_2 \\ r_3 \end{matrix}$$

$$a_1 \quad a_2 \quad a_3 \quad a_4$$

↳ $r_3 = r_2 - r_1$ but r_1, r_2 independent ⇝ $\dim(\underbrace{\operatorname{row}(A)}_{= \operatorname{span}\{r_1, r_2\}}) = 2$

$a_3 = a_1 - a_2$, $a_4 = 3a_1 - 2a_2$ ⇝ $\dim(\underbrace{\operatorname{col}(A)}_{\operatorname{span}\{a_3, a_4\}}) = 2$